



Validation of a Community-Acquired Pneumonia Score To Improve Empiric Antibiotic Selection at an Academic Medical Center

Meredith B. Oliver,^a Karen Fong,^b Laura Certain,^{c,d} Emily S. Spivak,^{c,d}  Tristan T. Timbrook^{e,f}

^aDepartment of Pharmacy, University of Minnesota Masonic Children's Hospital, Minneapolis, Minnesota, USA

^bDepartment of Pharmacy Services, University of Utah Health, Salt Lake City, Utah, USA

^cDepartment of Medicine, Division of Infectious Diseases, University of Utah School of Medicine, Salt Lake City, Utah, USA

^dVeterans Affairs Medical Center, Salt Lake City, Utah, USA

^eBiofire Diagnostics, Salt Lake City, Utah, USA

^fCollege of Pharmacy, University of Utah Health, Salt Lake City, Utah, USA

ABSTRACT The 2019 American Thoracic Society and the Infectious Diseases Society of America community-acquired pneumonia (CAP) guidelines recommend that drug-resistant pathogens (DRP) be empirically covered if locally validated risk factors are present. This retrospective case-control validation study evaluated the performance of the drug resistance in pneumonia (DRIP) clinical prediction score. Two hundred seventeen adult patients with ICD-10 (<https://www.who.int/classifications/classification-of-diseases>) pneumonia diagnosis, positive confirmed microbiologic data, and clinical signs and symptoms were included. A DRIP score of ≥ 4 was used to assess model performance. Logistic regression was used to select for significant predictors and create a modified DRIP score, which was evaluated to define clinical application. The DRIP score predicted pneumonia due to a DRP with a sensitivity of 67% and specificity of 73%. The area under the receiver operating characteristic (AUROC) curve was 0.76 (95% confidence interval [CI], 0.69 to 0.82). From regression analysis, prior infection with a DRP and antibiotics in the last 60 days, yielding scores of 2 points and 1 point, respectively, remained local risk factors in predicting drug-resistant pneumonia. Sensitivity (47%) and specificity (94%) were maximized at a threshold of ≥ 2 in the modified DRIP model. Therefore, prior infection with a DRP remained the only clinically relevant predictor for drug-resistant pneumonia. The original DRIP score demonstrated a decreased performance in our patient population and behaved similarly to other clinical prediction models. Empiric CAP therapy without antimethicillin-resistant *Staphylococcus aureus* and antipseudomonal coverage should be considered for noncritically ill patients without a drug resistant pathogen infection in the past year. Our data support the necessity of local validation to authenticate clinical risk predictors for drug-resistant pneumonia.

KEYWORDS pneumonia, bacterial, drug resistance, antimicrobial stewardship, validation study

Pneumonia accounts for more than 1.5 million hospital admissions each year in the United States, contributing to significant antibiotic use (1). The 2007, American Thoracic Society (ATS) and Infectious Diseases Society of America (IDSA) Community-Acquired Pneumonia (CAP) guideline endorsed the health care-associated pneumonia (HCAP) risk factors. When used, the HCAP criteria lead to an unnecessary 50% increase in broad-spectrum antimicrobial use in patients admitted with pneumonia (2, 3). Moreover, the true prevalence of CAP due to a drug-resistant pathogen (DRP) is likely

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Address correspondence to Tristan T. Timbrook, Tristan.Timbrook@utah.edu.

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TABLE 1 Patient characteristics^a

Characteristic	Value for patients with:		P value
	Non-DRPs (n = 117)	DRPs (n = 100)	
Age (yrs)	60 (52–70)	61 (48–69)	0.71
Male patients	67 (57)	56 (56)	0.85
Diabetes mellitus	27 (23.1)	31 (31)	0.19
Chronic pulmonary disease	55 (47)	48 (48)	0.88
Congestive heart failure	30 (25.6)	30 (30)	0.47
Chronic kidney disease	27 (23.1)	34 (34)	0.07
Cirrhosis	22 (18.8)	27 (27)	0.15
Malignancy	35 (29.9)	31 (31)	0.86
Transplant	15 (12.8)	27 (27)	0.01
Charlson comorbidity index	3 (0–7)	3 (0–8.25)	0.76
mEWS score	2 (1–4)	2 (1–4)	0.51
ICU admission	60 (51.3)	60 (60)	0.20
Mechanical ventilation	32 (27.4)	45 (45)	0.01
Vasopressor use	22 (18.8)	24 (24)	0.35
Length of stay (days)	5 (2–8)	6.5 (5–10)	0.001
Calculated DRIP score	2 (1–4)	5 (3–7)	<0.001
Patients with a DRIP score of ≥ 4	32 (27.4)	67 (67)	<0.001

^aData are presented as number (percentage) or median (interquartile range). Abbreviations: DRPs, drug-resistant pathogens; mEWS, modified early warning score; DRIP, drug resistance in pneumonia (score).

low (<2%) (4). Therefore, the 2019 joint ATS and IDSA CAP guidelines abandoned the HCAP score in favor of utilizing locally validated risk factors to identify patients at risk for CAP caused by methicillin-resistant *Staphylococcus aureus* (MRSA) or *Pseudomonas aeruginosa* (5).

The drug resistance in pneumonia (DRIP) clinical prediction score was developed to predict risk of pneumonia due to any DRP (6). After electronic support tool implementation at one of the DRIP derivation and validation sites, DRIP score utilization, coupled with MRSA nasal swab PCR-based de-escalation, reduced the use of empiric anti-MRSA and antipseudomonal agents, with no difference in length of stay, mortality, or cost (7). However, depending on institution-specific antimicrobial stewardship efforts, empiric antibiotic prescribing, and DRP rates, adopting the DRIP score at other institutions could lead to more broad-spectrum antibiotic use (8). External validation of this scoring system remains limited. To address these concerns and in concordance with ATS/IDSA guideline recommendations, we performed a retrospective case-control validation study to assess the utility of the DRIP clinical prediction score at a large academic medical center.

RESULTS

Two hundred seventeen adult patients were identified with an ICD-10 (<https://www.who.int/classifications/classification-of-diseases>) pneumonia diagnosis between 1 June 2018 and 31 December 2019 and included in this validation. Baseline characteristics are described in Table 1. The median patient age was 60 years (interquartile range [IQR], 51 to 70) for all patients included. More patients with pneumonia due to a DRP had a history of solid-organ or bone marrow transplant (27 [27.0%]) than did patients with pneumonia due to a non-DRP (15 [12.8%]). Patients infected with a DRP were more likely to receive mechanical ventilation (4 [45.0%]) than were those with non-DRP (32 [27.4%]). In addition, median length of stay was longer for patients with a DRP (6.5 days [IQR, 5 to 10]) than for those infected with a non-DRP (5 days [IQR, 2 to 8]). The median DRIP score was 5 (IQR, 3 to 7) in patients with a DRP, while the median DRIP score was 2 (IQR, 1 to 4) in patients without a DRP (Fig. 1). The most common organisms identified among DRP patients were *Pseudomonas aeruginosa* (50 [38.5%]) and MRSA (35 [26.9%]), while the most common among patients with a non-DRP were *Streptococcus pneumoniae* (43 [32.8%]) and methicillin-sensitive *S. aureus* (MSSA) (13

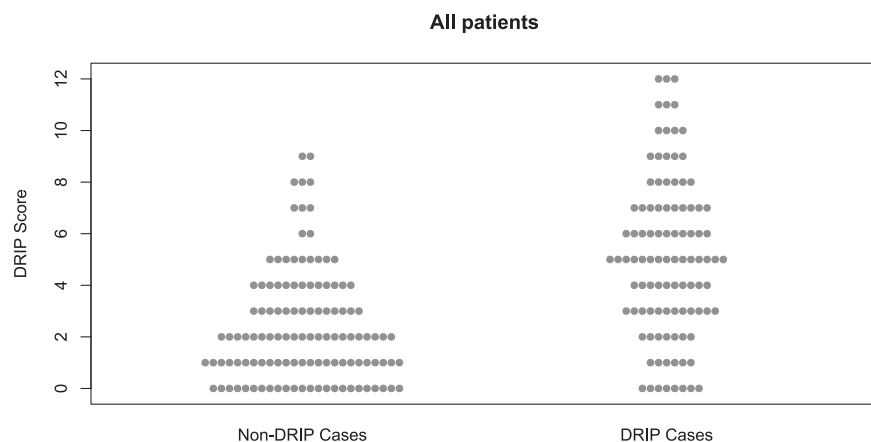


FIG 1 DRIP score distributions among patients with non-drug-resistant and drug-resistant pathogen cases.

[9.9%]) (Table 2). Many patients received anti-MRSA therapy (53 [45.3%]) and antipseudomonal therapy (63 [53.8%]) without identification of a DRP. In addition, many patients failed to receive anti-MRSA therapy (48 [48%]) and antipseudomonal therapy (34 [34%]) when microbiologic data confirmed a DRP. Median inpatient treatment durations were 9 days (IQR, 5.75 to 12) for patients with a DRP and 6 days (IQR, 4 to 9) for patients with non-DRP.

A DRIP score of ≥ 4 predicted pneumonia due to DRP at a sensitivity of 67% and specificity of 73% (Table 3). The area under the receiver operating characteristic (AUROC) curve was 0.76 (95% CI, 0.69 to 0.82). Sensitivity (57%) and specificity (84%) combined discrimination performance was maximized at a threshold of ≥ 5 in our validation. Stepwise logistic regression analysis excluded many risk factors from the origi-

TABLE 2 Microbiologic findings for the isolates in this study

Organism(s) ^a	No. (%) of isolates classified as non-DRP (n = 131)	No. (%) of isolates classified as DRP (n = 130)	P value
Normal respiratory flora	46 (35.1)	0	<0.001
<i>Streptococcus pneumoniae</i>	43 (32.8)	4 (3.1)	<0.001
MSSA	13 (9.9)	0	<0.001
MRSA	0	35 (26.9)	<0.001
Group B beta-hemolytic <i>Streptococcus</i>	1 (0.8)	0	0.99
<i>Corynebacterium striatum/simulans</i>	0	3 (2.3)	0.24
<i>Nocardia</i> spp.	0	3 (2.3)	0.24
<i>Escherichia coli</i>	8 (6.1)	4 (3.1)	0.38
<i>Klebsiella pneumoniae</i>	5 (3.8)	6 (4.6)	0.98
<i>Moraxella catarrhalis</i>	1 (0.8)	1 (0.8)	0.99
<i>Klebsiella oxytoca</i>	1 (0.8)	3 (2.3)	0.61
<i>Haemophilus influenzae</i>	8 (6.1)	3 (2.3)	0.22
<i>Klebsiella (Enterobacter) aerogenes</i>	1 (0.8)	1 (0.8)	0.99
<i>Enterobacter cloacae</i>	0	1 (0.8)	0.99
<i>Pseudomonas aeruginosa</i>	0	50 (38.4)	<0.001
<i>Citrobacter</i> spp.	1 (0.8)	1 (0.8)	0.99
<i>Serratia marcescens</i>	1 (0.8)	1 (0.8)	0.99
<i>Acinetobacter baumannii</i>	0	4 (3.1)	0.12
<i>Stenotrophomonas maltophilia</i>	0	6 (4.6)	0.03
<i>Morganella morganii</i>	0	1 (0.8)	0.99
<i>Achromobacter</i> spp.	0	3 (2.3)	0.24
<i>Legionella pneumophila</i>	1 (0.8)	0	0.99
Anaerobic Gram-negative rod	1 (0.8)	0	0.99

^aAbbreviations: MSSA, methicillin-sensitive *Staphylococcus aureus*; MRSA, methicillin-resistant *Staphylococcus aureus*.

TABLE 3 Sensitivity and specificity for each DRIP score threshold

DRIP score threshold	Sensitivity (%)	Specificity (%)
0	100	0
1	92	21
2	86	42
3	79	61
4	67	73
5	57	84
6	42	91
7	31	93
8	21	96
9	15	98
10	10	100
11	6	100
12	3	100
13	0	100
14	0	100

nal DRIP score model (Table 4). Antibiotic use within 60 days (AOR, 2.65 [95% CI, 1.42 to 4.97]; $P=0.002$) and prior infection with a DRP 1 year prior to admission (adjusted odds ratio [AOR], 10.6 [95% CI, 4.41 to 25.52]; $P<0.001$) remained significant risk factors in predicting drug-resistant pneumonia. Transforming the regression coefficients, the significant predictors of drug-resistant pneumonia yielded scores of 1 (antibiotics in 60 days) and 2 (prior infection with a DRP). These two predictors were used to create a modified DRIP model with a total score of 3. The modified DRIP score of 3 predicted pneumonia due to a DRP at a sensitivity of 39% and specificity of 95.8%. The AUROC of this modified DRIP model was 0.77 (95% CI, 0.71 to 0.83), which was not significantly different than the original DRIP AUROC ($P=0.48$). However, sensitivity (47%) and specificity (94%) combined discrimination performance was maximized at a threshold of ≥ 2 . In considering this modified DRIP threshold of ≥ 2 , prior infection with a DRP 1 year prior to admission remained the only clinically relevant predictor.

DISCUSSION

This single-center case-control validation study assessed the utility of the DRIP clinical prediction score in predicting pneumonia due to drug-resistant bacteria at a large academic medical center. The original DRIP score validation outperformed other pre-

TABLE 4 Risk factors for drug-resistant pneumonia

Risk factor ^a	Unadjusted			Adjusted			Score
	Odds ratio	95% CI	P value	Odds ratio	95% CI	P value	
Recent antibiotic use ^b	4.06	2.29–7.19	0.01	2.65	1.42–4.97	0.002	1
Long-term care resident ^c	1.6	0.67–3.82	0.15				
Tube feeds ^d	5.95	2.15–16.46	0.22				
Prior infection with DRP ^e	13.94	5.9–32.9	<0.001	10.6	4.41–25.52	<0.001	2
Recent hospitalization ^b	1.84	1.05–3.22	0.95				
Chronic lung disease	1.04	0.61–1.78	0.13				
Poor functional status ^f	2.35	1.29–4.28	0.13				
Gastric acid suppression ^g	2.43	1.41–4.2	0.14				
Wound care ^h	2.84	1.27–6.38	0.78				
MRSA colonization ^e	4.27	1.5–12.11	0.24				

^aVariance inflation factor (VIF) of <3 for all 10 variables.

^bWithin 60 days of admission.

^cWithin 180 days of admission; excludes assisted living or group home facility.

^dActive at time of admission.

^eWithin 1 year of admission at any site.

^fKarnofsky score of <70 or nonambulatory status at time of admission.

^gH2 blocker or proton pump inhibitor within 14 days of admission.

^hWithin 30 days of admission.

diction models, with an AUROC of 0.88 (95% CI, 0.82 to 0.93) (6). However, the original DRIP score demonstrated a decreased performance (AUROC, 0.76 [95% CI, 0.69 to 0.82]) at our institution. While our study did not seek to compare our observed DRIP score performance with other clinical prediction models, our results do suggest that it behaves similarly to other pneumonia DRP prediction scores in the literature (6, 9–14). In addition, we found a DRIP score of ≥ 5 to demonstrate the best balance of sensitivity and specificity in our population, as opposed to ≥ 4 in the original validation model (6).

Our finding of decreased AUROC compared to that of the original validation study may be due to real-world DRIP score performance and retrospective nature of our validation case-control study. External validation studies for prediction models are rarely performed yet are arguably needed to address their generalizability and clinical applicability outside the original setting developed (15, 16). In the original DRIP score derivation and validation study, risk predictors were confirmed with the patient via telephone if not available in the electronic health record (EHR) (6). In any retrospective validation, confirming the accuracy of these predictors is challenging. However, and perhaps more importantly, confirming the presence and accuracy of these predictors for every patient upon emergency room presentation is unlikely and lacks feasibility. Therefore, our validation provides a more accurate, real-world depiction of DRIP score performance. We encourage additional local validation studies to further authenticate clinical risk predictors for drug-resistant pneumonia.

The DRIP score was independently assessed at a Veterans Affairs hospital, and it was concluded that empiric broad-spectrum antimicrobial use would increase by 9% if the DRIP score was adopted (8). They attribute their low empiric broad-spectrum antibiotic use to active antimicrobial stewardship efforts, low rates of DRP, and local culture that discouraged use of HCAP criteria (8).

Prediction score implementation can also be difficult due to barriers for adoption and utilization. Clinical workflow integration can be challenging, especially when clinical prediction scores include large numbers of risk factors that are difficult to obtain from an electronic health record (17). Moreover, prediction models have been reported in survey data of physicians as time-consuming, difficult to use, and not always providing value (18). Given these challenges, solutions for implementation are required. Our modified DRIP score considers only prior DRP infection within 1 year, offering clinicians a quick heuristic for assessment and easy adoption. A similar heuristic approach using previous culture results as a stewardship intervention to improve initial empiric antibiotics has been reported recently, reflecting improved prescribing (19). Further exploration of such approaches should be evaluated to maximize net benefit in reducing both under- and overprescribing.

Our study has several limitations. ICD-10 codes were used to derive our cohort. However, pneumonia diagnostic codes have been shown to be valid, and potentially superior to clinical symptoms and radiologic evidence, in identifying medical record notations of pneumonia (20). To further offset this limitation, we confirmed ICD-10 pneumonia cases with supporting clinical signs and symptoms, chest imaging, and microbiologic diagnosis. While microbiology results of organisms identified in sputum may be inaccurate in correlating to pneumonia due to detection of colonizing species and poor specimens, specimen quality standards were in place for the study: our laboratory rejects sputum samples when ≥ 10 epithelial cells per low-power field or more epithelial cells than polymorphonuclear leukocytes (PMN) are identified. In addition, we recommend continued external validation of drug-resistant CAP predictor models prior to implementation, given that our single-center results may not be generalizable to other institutions. Evaluating incidence was not a primary goal; we employed a case-control study design for efficiency and feasibility, thus failing to capture the true incidence of DRP at our institution. We therefore could not capture DRIP score or modified DRIP score impact on antibiotic prescribing. Given our low rate of DRPs, we plan

to assess the impact of any prediction score on broad-spectrum antibiotic use prior to implementation.

Conclusion. Locally validated clinical predictors can serve as an antimicrobial stewardship tool to decrease empiric broad-spectrum antimicrobial use. This external validation study at an academic medical center evaluated DRIP score performance (AUROC, 0.76), which suggests a performance similar to those of other reported clinical prediction models in the literature for drug-resistant pneumonia. Given the performance and challenges of DRIP score use, a heuristic approach may be to give empiric CAP therapy without anti-MRSA or anti-pseudomonal coverage for non-critically ill patients without a 1-year-prior infection with a drug-resistant pathogen. We recommend additional local validation studies to further authenticate clinical risk predictors for drug-resistant pneumonia.

MATERIALS AND METHODS

Adult patients admitted to an academic medical center from the period between 1 June 2018 and 31 December 2019 were initially selected by ICD-10 pneumonia diagnosis and confirmed by a single manual chart reviewer for positive microbiologic data obtained within 48 h of admission, signs of infection (i.e., leukocytosis, fever, cough, or sputum production), and abnormal chest radiography. The DRIP score includes two points for antibiotic use within 60 days, long-term care resident (not including assisted living or group home facility) in last 180 days, active tube feeds at time of admission, and prior infection with a DRP within 1 year of admission and one point for hospitalization within 60 days, chronic lung disease, poor functional status at time of admission (Karnofsky score of <70 or nonambulatory status), gastric acid suppression (H2 blocker or proton pump inhibitor) within 14 days of admission, recipient of outpatient wound care within 30 days of admission, and MRSA colonization or infection with MRSA within 1 year of admission (4, 6, 21, 22).

A microbiologic diagnosis was confirmed if an organism consistent with a bacterial respiratory pathogen was identified on blood culture, sputum culture, pleural fluid culture, or urinary antigen test. Patients were excluded if microbiologic diagnostic studies were positive for coagulase-negative staphylococci, *Micrococcus* spp., *Neisseria* spp., *Enterococcus* spp., diphtheroids, non-*Streptococcus pneumoniae* alpha-hemolytic streptococci, or *Mycobacterium* spp., as well as viral or fungal etiologies. Patients with cystic fibrosis or pregnancy or those transferred from an outside hospital after 24 h of health care contact were also excluded. As a validation case-control study, cases were collected consecutively until at least 100 cases of pneumonia with DRP and 100 controls of pneumonia without DRP were achieved (15).

Microbiologic data were collected from the following sources: sputum sample, tracheal aspirate, bronchoalveolar lavage (BAL) fluid, or pleural fluid, blood culture, or urinary antigen tests for *Legionella* spp. and *Streptococcus pneumoniae*. Organisms were classified as a DRP if the patient's cultures demonstrated resistance to 2019 ATS/IDSA CAP guideline-recommended antimicrobials (ceftriaxone, macrolide, or fluoroquinolone) or the organism was identified as MRSA, *Pseudomonas aeruginosa*, an *Acinetobacter* sp., or a *Stenotrophomonas* sp. (5, 6). A positive *S. pneumoniae* urinary antigen was classified as a non-DRP, since ceftriaxone and levofloxacin resistances of *S. pneumoniae* at our institution remain low ($<5\%$).

Intensive care unit (ICU) admission was defined as an admission to the ICU within 48 h of hospital presentation. Mechanical ventilation was defined as any positive-pressure ventilation (bilevel positive airway pressure [BiPAP], continuous positive airway pressure [CPAP], or mechanical ventilation) within 48 h of admission. Vasopressor use was defined by the use of norepinephrine, vasopressin, epinephrine, dopamine, phenylephrine (intravenous [i.v.]), milrinone, or isoproterenol within 48 h of admission. Transplant patients included those with a history of solid-organ or bone marrow transplant.

Antibiotics providing coverage against MRSA (vancomycin and linezolid) and *Pseudomonas aeruginosa* (piperacillin-tazobactam, cefepime, ceftazidime, imipenem, meropenem, and aztreonam, excluding fluoroquinolones) were defined as broad spectrum (5). The following antibiotics were not characterized as broad spectrum: ampicillin/sulbactam, ceftriaxone, azithromycin, clarithromycin, and fluoroquinolones (6). These definitions are consistent with the 2019 ATS/IDSA CAP guidelines.

Statistical analysis. Demographic data were summarized using median (IQR) and number (percentage). The primary outcome assessed performance of DRIP and associated measures (including sensitivity, specificity, and area under the receiver operating characteristic curve [AUROC]). The secondary outcome included an evaluation of a modified DRIP score. Unconditional logistic regression was used to model DRP risk. Multicollinearity was assessed with variance inflation factor to ensure that the strength of each DRIP score variable was independent from all others in the model. Significant predictors were chosen by manual backward stepwise selection, iteratively removing the most nonsignificant predictor, to create a modified DRIP score. DRIP and modified DRIP model AUROCs were compared using the DeLong test (23). All analyses were performed in R version 3.5.0 (R Foundation for Statistical Computing, Vienna, Austria).

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